*Sustainable transformation through innovation"*

**SCHOOL OF SCIENCE AND TECHNOLOGY**

**BACHELOR OF COMPUTER SCIENCE**

**BSc. Comp 122 – SOFTWARE ENGINEERING**

**BSc. Comp 122 - SOFTWARE ENGINEERING**

**BACKGROUND AND RATIONALE**

Software Engineering module introduces learners to software engineering. The modules begins with explaining key concepts in software engineering, explains the concept of tools and environments, explains the process of project management, explains software process models, explains the process of software analysis and design using UML, explains software requirements, analysis, validation and management. The module further explains software design, design concepts, design patterns. It also explain software testing, testing approaches, testing levels (unit, integration, system, acceptance), it explains software analysis, program optimisation and correctness, static and dynamic analysis, explains software maintenance, corrective, adaptive, perfective and preventive maintenance, explains software quality management, assurance, planning, control, and ends with explaining agile software development, methods and practices.

**LEARNING OUTCOMES**

**At the end of this module learners should be able to:**

- Know and apply key concepts of the analysis, design, development and maintenance of complex software systems and infrastructures.
- Know and apply knowledge on the software lifecycle and its stages and utilise these in the development of software process models.
- Know and knowledge in conducting software tests in order to evaluate and verify software products.
- Know and apply the knowledge in software analysis, program optimisation and correctness, static and dynamic analysis
- Know and apply the knowledge in software maintenance, corrective, adaptive, perfective and preventive maintenance
- Know and apply knowledge on software quality management, assurance, planning, control
- Know and apply the knowledge in agile software development methods and practices.

**COURSE CONTENTS**

-UNIT 1: Key concepts in Software Engineering.

UNIT 2: Tools and Environments.

UNIT 3: Project Management.

UNIT 4: Software Process Models.

UNIT5: Software Analysis and Design using UML.

UNIT 6: Software Requirements, analysis, validation and management.

UNIT 7: Software Design, design concepts, design patterns.

UNIT 8: Software Testing, testing approaches, testing levels (unit, integration, system, acceptance).

UNIT 9: Software Analysis, program optimisation and correctness, static and dynamic analysis.

UNIT 10: Software Maintenance, corrective, adaptive, perfective and preventive maintenance.

UNIT 11: Software Quality Management, assurance, planning, control.

UNIT 12: Agile Software Development, methods and practices.

**ASSESSMENT**

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|---|---|
| Assignment 1 | 15% |
| Assignment 2 | 15% |
| Final exam | 70% |
| Final | 100% |

**PRESCRIBED READING:**

Lethbridge T.C and Laganiere R. (2005) Object-Oriented Software Engineering: Practical software development using UML and Java. Berkshire: McGraw Hill Education

Olga Filipova and Rui Vilão (2018). Software Development from A to Z: A Deep Dive into all the Roles Involved in the Creation of Software. Berlin: Filipova and Vilao

Michael Keeling (2017). Design It!: From Programmer to Software Architect. North Carolina: The Pragmatic Programmer

**RECOMMENDED READING:**

Richard F. Schmidt (2013). Software Engineering Architecture-Driven Software Development. London: Elsevier Inc.

Humberto Cervantes and Rick Kazman. (2016) Designing Software Architectures: A Practical Approach. Boston: Addison-Wesley